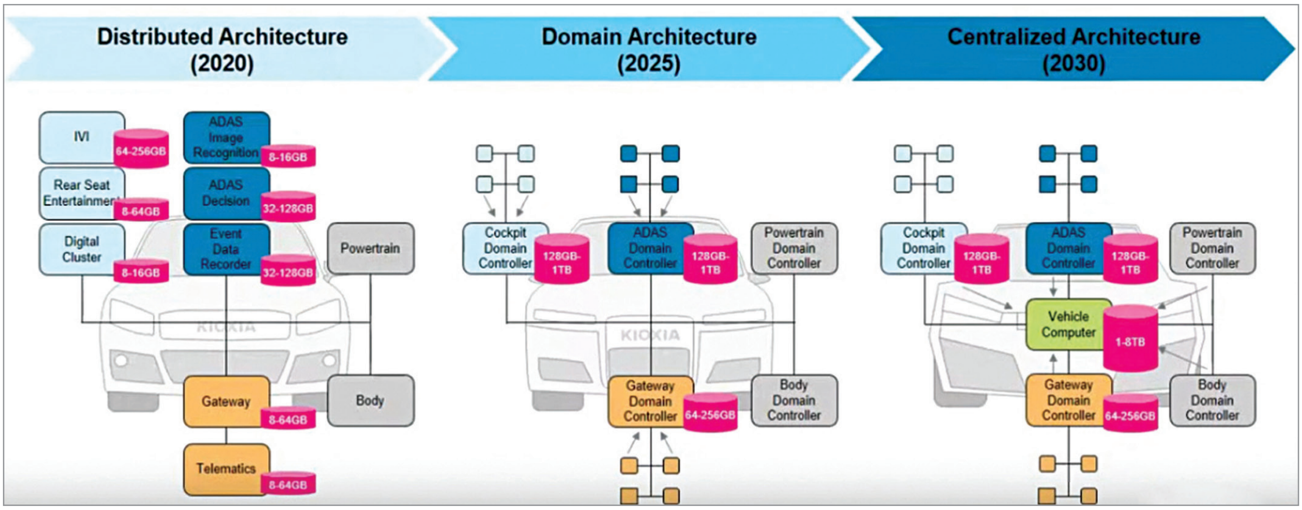

Fig. 3: Evolution of the automotive system (Credit: KIOXIA)

Different flavours of SLC NAND Flash Memory

- ***KIOXIA.*** A Toshiba spin-out, KIOXIA offers memory solutions such as SLC NAND, BENAND and Serial Interface NAND. Except for Serial Interface NAND that uses SPI driver for the host controller interface, the other two use NAND driver for the same. While the BENAND and Serial Interface NAND have built-in error code correction (ECC), the SLC NAND does not.
- ***Micron Technology.*** Its products address growing opportunities for memory and storage hierarchy innovation. Products include 3D NAND, TLC NAND, MLC NAND, and SLC NAND.
- ***Cypress Semiconductor.*** It has a vast range of RoHS-compliant NAND flash memory ICs.

No matter the company, they all come with page sizes ranging from (2048+64)x8 to (4096+256)x8 and block sizes of (128k+4k)x8 to (256k+16k)x8 across bit sizes of 1Gbit, 2Gbit, and 4Gbit. They also perform bad block management, wear levelling, and garbage collection.

phones, digital cameras, and USB flash drives.

Future automotive requirements

The increased enhancement in embedded systems is rapidly transforming the automotive sector. The next-generation vehicles, particularly autonomous vehicles, rely on high computing power for managing electrical/electronic (E/E) architecture that includes complex embedded control systems and related hardware/software.

Over here, both NAND and NOR—along with EEPROM, DRAM, and FRAM—play a significant role in automotive E/E systems. In 2020, the distributed architecture, which included ADAS image recognition, ADAS detection, and event data recorder was introduced. On the infotainment side, there was in-vehicle infotainment (IVI), rear seat entertainment, and digital cluster. For the gateway, there was Gateway and Telematics. Today, up to 256GB for infotainment and 16GB or 32GB for safety and gateway systems can be achieved.

With enhanced centralised storage and processing capabilities, the year 2025 will see distributed architecture turn into domain architecture with all the parts/blocks getting integrated. So, the IVI and rear-seat entertainment system, along with other infotainment systems, will get under the cockpit domain controller, requiring up to 1TB.

The safety system will become the ADAS domain controller that will include all the recognition and detection, even the data recorder. It will also be up to 1TB. The gateway domain controller, including telematics, may need more than 256GB. Powertrain and body domain controllers will not require much storage.

By 2030, all the above are expected to move to a centralised architecture, which will be controlled by vehicle's computer, a high-speed processor requiring 1TB to 8TB. It will also control the cockpit domain, safety domain, and gateway domain. Together, they will require 10TB or more. Obviously, it would be a drastic increase in the storage requirements for the vehicle of the future.

Outcome

After going through the above details, a question comes to mind. Should all NOR devices be changed to NAND devices? Well, it all boils down to the application. NOR can efficiently handle boot speeds as it has better data reading capability. But where NOR fails at adequate data storage, NAND can step in as it has better write speed. EFY

The article is based on a talk by Ong Chuen Wee, Engineering Manager at Memory Marketing Department, KIOXIA Singapore. It has been transcribed and curated by Vinay Prabhakar Minj, an IoT and audio electronics enthusiast.